

Lecture 4: Convex Optimization

1 What Makes an Optimization Problem Convex?

Consider

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{subject to } x \in X.$$

The problem is convex when the objective f is convex and the feasible set X is convex. Thus, understanding convex optimization reduces largely to recognizing convex functions and convex sets.

2 The Chord Test and the Epigraph

Definition 1 (Convex function). *A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex if*

$$f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y), \quad \alpha \in [0, 1].$$

Geometrically, the graph of a convex function lies below every chord joining two points on the graph. The epigraph is

$$\text{epi } f = \{(x, t) : f(x) \leq t\}.$$

A function is convex exactly when its epigraph is a convex set.

Why this is true (intuition). Take two points $(x, f(x))$ and $(y, f(y))$ on the boundary of the epigraph. A point on the line segment between them is

$$(\alpha x + (1 - \alpha)y, \alpha f(x) + (1 - \alpha)f(y)).$$

For the whole segment to stay in the epigraph, its height must be at least the function value at the interpolated input. This is precisely the convexity inequality.

3 Two Practical Convexity Tests

3.1 Convexity along all lines

For any $x_0, h \in \mathbb{R}^n$, define

$$\phi(t) = f(x_0 + th).$$

Then f is convex if and only if ϕ is convex for every choice of x_0 and h .

Why this is true (intuition). Convexity compares a function only along line segments. Any line segment from x to y can be parameterized by $x + t(y - x)$. Therefore, if the function bends upward along every possible line, it is convex in the full space. Conversely, restricting a convex function to any line preserves the same chord inequality.

3.2 Positive semidefinite Hessian

If f is twice differentiable, then

$$f \text{ is convex} \iff \nabla^2 f(x) \succeq 0 \quad \forall x.$$

Why this is true (intuition). Along a direction h , let $\phi(t) = f(x + th)$. Its second derivative is

$$\phi''(t) = h^\top \nabla^2 f(x + th) h.$$

If the Hessian is PSD, then $h^\top \nabla^2 f h \geq 0$ for every direction h . Hence every one-dimensional slice has nonnegative second derivative and bends upward. By the line test, f is convex. The reverse direction follows in the same way: if every line slice is convex, then every directional second derivative is nonnegative.

4 Scalar, Affine, and Quadratic Examples

For scalar functions, the second derivative gives an immediate test:

- $f(x) = x^2$: $f''(x) = 2 > 0$, so it is strictly convex on $\mathbb{R}$.
- $f(x) = -\log x$: $f''(x) = 1/x^2 > 0$, so it is strictly convex on $x > 0$.
- $f(x) = 1/x$: $f''(x) = 2/x^3 > 0$, so it is strictly convex on $x > 0$.

Domain matters. Convexity is stated on a convex domain. In particular, the claims for $-\log x$ and $1/x$ above use the domain $x > 0$.

An affine function

$$f(x) = a^\top x + b$$

has $\nabla^2 f(x) = 0$, so it is both convex and concave.

For a quadratic function

$$f(x) = \frac{1}{2} x^\top Q x + b^\top x + c, \quad Q = Q^\top,$$

we have $\nabla^2 f(x) = Q$. Therefore,

$$f \text{ is convex} \iff Q \succeq 0.$$

This is the multidimensional version of a bowl-shaped quadratic.

5 Norms Are Convex

A norm satisfies non-negativity, homogeneity, and the triangle inequality:

$$\|x + y\| \leq \|x\| + \|y\|, \quad \|\alpha x\| = |\alpha| \|x\|.$$

For $\alpha \in [0, 1]$,

$$\begin{aligned} \|\alpha x + (1 - \alpha)y\| &\leq \|\alpha x\| + \|(1 - \alpha)y\| \\ &= \alpha \|x\| + (1 - \alpha) \|y\|. \end{aligned}$$

Hence every norm is convex.

The ℓ_q norm is

$$\|x\|_q = \left(\sum_{i=1}^n |x_i|^q \right)^{1/q}, \quad q \geq 1.$$

The notation $\|x\|_0 = \#\{i : x_i \neq 0\}$ counts nonzero entries, but it is not a norm.

The unit balls illustrate different geometries: the ℓ_1 ball is associated with sparsity, the ℓ_2 ball has rotational symmetry, and the ℓ_∞ ball imposes coordinate-wise bounds.

6 Closure Rules for Convex Functions

Closure rules allow us to recognize complicated convex functions from simpler ones.

6.1 Nonnegative scaling and sums

If f is convex and $\alpha \geq 0$, then αf is convex. More generally, if each f_i is convex and $w_i \geq 0$, then

$$\sum_i w_i f_i(x)$$

is convex.

Why this is true (intuition). Start with the convexity inequality for each f_i . Multiplying an inequality by a nonnegative number preserves its direction, and adding valid inequalities preserves validity. This immediately gives the convexity inequality for the weighted sum.

6.2 Affine precomposition

If f is convex, then

$$g(x) = f(Ax + b)$$

is convex.

Why this is true (intuition). Affine maps preserve weighted averages:

$$A(\alpha x + (1 - \alpha)y) + b = \alpha(Ax + b) + (1 - \alpha)(Ay + b).$$

Applying convexity of f to the right-hand side gives the desired inequality for g .

6.3 Pointwise supremum

If every f_β is convex, then

$$f(x) = \sup_{\beta \in \mathcal{A}} f_\beta(x)$$

is convex. For example, $|x| = \max\{x, -x\}$.

Why this is true (intuition). For every β ,

$$f_\beta(\alpha x + (1 - \alpha)y) \leq \alpha f_\beta(x) + (1 - \alpha)f_\beta(y) \leq \alpha f(x) + (1 - \alpha)f(y).$$

Since every candidate f_β is below the same upper bound, their supremum is also below it.

7 Composition Rules

Let $f(x) = h(g(x))$. Important sufficient conditions for convexity are:

- g convex, and h convex and nondecreasing;
- g concave, and h convex and nonincreasing;
- g affine, and h convex.

Why monotonicity matters (intuition).

Second-order explanation. Assume that g and h are twice differentiable. For

$$f(x) = h(g(x)),$$

the gradient is

$$\nabla f(x) = h'(g(x))\nabla g(x).$$

Differentiating once more gives

$$\boxed{\nabla^2 f(x) = h''(g(x))\nabla g(x)\nabla g(x)^T + h'(g(x))\nabla^2 g(x)}.$$

Recall that a twice differentiable function is convex if its Hessian is positive semidefinite. Therefore, we only need to determine when the two terms above are positive semidefinite.

Case 1: g convex, h convex and nondecreasing. Since h is convex,

$$h''(g(x)) \geq 0.$$

Also,

$$\nabla g(x)\nabla g(x)^T \succeq 0,$$

because for any vector v ,

$$v^T \nabla g(x)\nabla g(x)^T v = \left(\nabla g(x)^T v\right)^2 \geq 0.$$

Hence,

$$h''(g(x))\nabla g(x)\nabla g(x)^T \succeq 0.$$

Since h is nondecreasing,

$$h'(g(x)) \geq 0,$$

and since g is convex,

$$\nabla^2 g(x) \succeq 0.$$

Therefore,

$$h'(g(x))\nabla^2 g(x) \succeq 0.$$

Both terms are positive semidefinite, so

$$\boxed{\nabla^2 f(x) \succeq 0}.$$

Thus $f(x) = h(g(x))$ is convex.

Case 2: g concave, h convex and nonincreasing. Again, convexity of h gives

$$h''(g(x)) \geq 0,$$

so the first term is positive semidefinite.

Now,

$$h'(g(x)) \leq 0$$

because h is nonincreasing, while

$$\nabla^2 g(x) \preceq 0$$

because g is concave. Therefore,

$$h'(g(x))\nabla^2 g(x) \succeq 0.$$

Hence,

$$\boxed{\nabla^2 f(x) \succeq 0},$$

and f is convex.

Case 3: g affine and h convex. If

$$g(x) = a^T x + b,$$

then

$$\nabla^2 g(x) = 0.$$

Thus

$$\nabla^2 f(x) = h''(g(x))aa^T.$$

Since h is convex, $h''(g(x)) \geq 0$, and $aa^T \succeq 0$. Therefore,

$$\boxed{\nabla^2 f(x) \succeq 0},$$

so f is convex.

Examples. For

$$f(x) = e^{g(x)},$$

the outer function $h(t) = e^t$ satisfies

$$h'(t) = e^t > 0, \quad h''(t) = e^t > 0.$$

Therefore, if g is convex, then $e^{g(x)}$ is convex.

Similarly, for

$$f(x) = -\log(g(x)),$$

the outer function $h(t) = -\log t$, $t > 0$, satisfies

$$h'(t) = -\frac{1}{t} < 0, \quad h''(t) = \frac{1}{t^2} > 0.$$

Therefore, if $g(x) > 0$ is concave, then $-\log(g(x))$ is convex.

8 Convex Functions of Matrices

Several matrix-valued examples can be recognized using the same closure rules.

The affine trace function

$$f(X) = \text{Tr}(A^\top X) + b$$

is affine and hence convex.

For a symmetric PSD matrix X ,

$$\lambda_{\max}(X) = \sup_{\|u\|_2=1} u^\top X u.$$

For each fixed u , $u^\top X u = \text{Tr}(u u^\top X)$ is affine in X . Therefore $\lambda_{\max}$ is a pointwise supremum of affine functions and is convex.

Similarly,

$$\sigma_1(X) = \|X\|_2 = \sup_{\|u\|=\|v\|=1} u^\top X v$$

is convex because each $u^\top X v$ is affine in X and the supremum preserves convexity.

The Ky Fan norms are

$$\|X\|_{(k)} = \sum_{i=1}^k \sigma_i(X),$$

and for $k = \min\{m, n\}$ this becomes the nuclear norm.

9 From Convex Functions to Convex Sets

9.1 Sublevel sets

For a convex function f , the sublevel set

$$C_\alpha = \{x \in \text{dom } f : f(x) \leq \alpha\}$$

is convex. The converse is not generally true.

Why this is true (intuition). If $x, y \in C_\alpha$, then $f(x) \leq \alpha$ and $f(y) \leq \alpha$. Convexity gives

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y) \leq \alpha.$$

Thus the entire segment between x and y remains in the same sublevel set.

Examples of convex sets include affine sets $\{x : Ax = b\}$, halfspaces $\{x : a^\top x \leq b\}$, and norm balls $\{x : \|x - c\| \leq r\}$.

9.2 Intersections and polyhedra

An arbitrary intersection of convex sets is convex:

$$\bigcap_{\alpha \in \mathcal{A}} C_\alpha.$$

Why this is true (intuition). If two points lie in the intersection, they lie in every individual C_α . Since each C_α contains the line segment between them, that segment lies in every set simultaneously, and hence in the intersection.

A polyhedron

$$\{x : Ax \leq b\} = \bigcap_i \{x : a_i^\top x \leq b_i\}$$

is therefore convex. In contrast, a union of convex sets is generally not convex.

10 Convex Cones

A convex cone $C \subseteq \mathbb{R}^n$ is a convex set that also satisfies

$$x \in C, \alpha \geq 0 \implies \alpha x \in C.$$

Examples include the nonnegative orthant $\mathbb{R}_+^n$, the quadratic cone

$$\{(x, t) : \|x\| \leq t\},$$

and the PSD cone

$$\{X : X = X^\top \succeq 0\}.$$

11 Affine Images and Preimages

Let $T(x) = Ax + b$ and let C be convex. Then both

$$T(C) = \{Ax + b : x \in C\}$$

and

$$T^{-1}(C) = \{x : Ax + b \in C\}$$

are convex.

Why this is true (intuition). Affine transformations preserve convex combinations. For the image, take $T(x_1)$ and $T(x_2)$; their convex combination equals $T(\alpha x_1 + (1 - \alpha)x_2)$, and the input remains in C . For the preimage, if $Ax_1 + b$ and $Ax_2 + b$ lie in C , then their convex combination lies in C , and that combination equals $A(\alpha x_1 + (1 - \alpha)x_2) + b$.

An important example is the feasible set of a linear matrix inequality,

$$\left\{ x : A_0 + \sum_{i=1}^n x_i A_i \succeq 0 \right\}, \quad A_i = A_i^\top,$$

which is an affine preimage of the PSD cone and is therefore convex.

11.1 A Convex Problem with an Affine Equality Constraint

12 Homework 1 Extensions: Applications of Convex Optimization

The following examples come from Homework 1 and extend the lecture material. They are included as applications rather than as lecture theorems, so the results are summarized without adding full proofs.

Homework 1 considers

$$\begin{aligned} \min_x \quad & \sum_{i=1}^n e^{x_i} \\ \text{subject to} \quad & \sum_{i=1}^n x_i = \gamma. \end{aligned}$$

This is a convex optimization problem because each e^{x_i} is convex, their sum is convex, and the equality constraint defines an affine (hence convex) feasible set.

The equality constraint can be eliminated using

$$x_n = \gamma - \sum_{i=1}^{n-1} x_i,$$

which gives the unconstrained problem

$$\min_{x_1, \dots, x_{n-1}} \left[\sum_{i=1}^{n-1} e^{x_i} + e^{\gamma - \sum_{i=1}^{n-1} x_i} \right].$$

At the optimum, symmetry and the first-order conditions give

$$x_1 = \dots = x_n = \frac{\gamma}{n}.$$

Thus the unique global minimizer is

$$x^* = \frac{\gamma}{n} \mathbf{1}.$$

12.0.1 Connection to the arithmetic–geometric mean inequality

With the change of variables $y_i = e^{x_i} > 0$, we have

$$\prod_{i=1}^n y_i = e^{\sum_{i=1}^n x_i} = e^{\gamma}.$$

The optimality result above implies

$$\sum_{i=1}^n e^{x_i} \geq n e^{\gamma/n}.$$

Substituting $y_i = e^{x_i}$ and $\gamma = \sum_i \log y_i$ gives

$$\frac{1}{n} \sum_{i=1}^n y_i \geq \left(\prod_{i=1}^n y_i \right)^{1/n}.$$

This is the arithmetic–geometric mean inequality. The optimization viewpoint says that, for a fixed product, the sum is minimized when all positive variables are equal.

12.1 Log-Sum-Exp

Homework 1 uses the function

$$f(x) = \log \left(\sum_{i=1}^n e^{a_i^\top x + b_i} \right),$$

called the *log-sum-exp* function. It is convex and appears frequently as a smooth approximation to a maximum.

If

$$p_i(x) = \frac{e^{a_i^\top x + b_i}}{\sum_j e^{a_j^\top x + b_j}}, \quad \sum_i p_i = 1,$$

then

$$\nabla f(x) = \sum_i p_i a_i$$

and

$$\nabla^2 f(x) = \sum_i p_i a_i a_i^\top - \left(\sum_i p_i a_i \right) \left(\sum_i p_i a_i \right)^\top \succeq 0.$$

A useful interpretation is that the Hessian has the form of a weighted covariance matrix of the vectors a_i , which explains why it is PSD.

Extension. Log-sum-exp behaves like a smooth maximum. If $z_i = a_i^\top x + b_i$, then

$$\max_i z_i \leq \log \sum_i e^{z_i} \leq \max_i z_i + \log n.$$

12.2 Largest Singular Value and Composition

The homework considers

$$f(X) = \exp \left((\sigma_1(AX))^{10} \right),$$

where $\sigma_1(Z)$ is the largest singular value of Z .

The convexity can be organized using the closure rules from Lecture 4:

1. $X \mapsto AX$ is affine.
2. $Z \mapsto \sigma_1(Z) = \|Z\|_2$ is a norm and hence convex and nonnegative.
3. $t \mapsto t^{10}$ is convex and nondecreasing on $t \geq 0$.
4. $s \mapsto e^s$ is convex and increasing.

Therefore the successive compositions preserve convexity.

12.3 Parameterized Quadratic Objectives

Another homework problem studies

$$f(x, y) = x^2 + y^2 + \beta xy + x + 2y.$$

Its Hessian is

$$\nabla^2 f = \begin{bmatrix} 2 & \beta \\ \beta & 2 \end{bmatrix},$$

whose eigenvalues are

$$2 + \beta, \quad 2 - \beta.$$

Hence

$$f \text{ is convex} \iff |\beta| \leq 2, \quad f \text{ is strictly convex} \iff |\beta| < 2.$$

This example is useful because it shows how a single parameter can change the geometry of an optimization problem from convex to nonconvex.

For $\beta \neq \pm 2$, the stationary point obtained from $\nabla f = 0$ is

$$x^* = \frac{2\beta - 2}{4 - \beta^2}, \quad y^* = \frac{\beta - 4}{4 - \beta^2}.$$

When $|\beta| < 2$, strict convexity makes this stationary point the unique global minimizer. At the boundary values $\beta = \pm 2$, the Hessian is PSD but singular, so the linear terms must be checked separately for existence of a minimizer. When $|\beta| > 2$, the Hessian is indefinite and the stationary point is not a global minimum.

12.4 Linewise Information Is Not Enough in General

Homework 1 also emphasizes a useful distinction. If x^* is a local minimum of a differentiable function along every line through x^* , then

$$\nabla f(x^*) = 0.$$

Indeed, for every direction d , the one-dimensional function

$$g(\alpha) = f(x^* + \alpha d)$$

has $g'(0) = \nabla f(x^*)^\top d = 0$. Since this holds for every d , the gradient must be zero.

However, for a general nonconvex function, being a local minimum along every fixed line does *not* guarantee being a local minimum in the full space. The homework counterexample

$$f(y, z) = (z - py^2)(z - qy^2), \quad 0 < p < q,$$

shows that a curved path can reveal descent that no fixed line reveals.

Connection to convexity. For convex differentiable functions, the situation is much stronger: $\nabla f(x^*) = 0$ is sufficient for global optimality. The counterexample above matters because this guarantee disappears outside convex optimization.

12.5 Nonconvex Matrix Factorization as a Contrast

The last homework problems consider objectives such as

$$\frac{1}{2} \|A - xx^\top\|_F^2 \quad \text{and} \quad \frac{1}{2} \|A - xy^\top\|_F^2.$$

These problems are not convex in their optimization variables. They are useful contrasts to the convex theory: one cannot conclude that every stationary point is globally optimal merely from first-order conditions.

At the same time, these structured nonconvex problems can have special landscapes in which all *local minima* are global minima. That property is problem-specific; it should not be confused with the general convex-optimization theorem that every local minimum of a convex problem is global.